

System Documentation

Electronic Control System

BlueVision | NewGeneration

Interface to non-MTU remote control system

E532815/00E



Power. Passion. Partnership.

Printed in Germany

© 2014 Copyright MTU Friedrichshafen GmbH

This Publication is protected by copyright and may not be used in any way whether in whole or in part without the prior written permission of MTU Friedrichshafen GmbH. This restriction also applies to copyright, distribution, translation, microfilming and storage or processing on electronic systems including data bases and online services.

This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

Table of Contents

1	Safety		2.3	Analog outputs and binary outputs	17
			2.3.1	Analog outputs	17
			2.3.2	Binary outputs	18
1.1	Important provisions for all products	4			
1.2	Personnel and organizational requirements	5			
1.3	Safety regulations for startup and operation	6			
1.4	Safety regulations for maintenance and repair work	7	3	Installation and Commissioning Instructions	
1.5	Standards for safety notices in the text	10	3.1	Connecting a non-MTU remote control system (shipyard supply) to LOP	20
2	Functional Description		4	Appendix A	
2.1	Interfaces on LOP to non-MTU remote control systems	11	4.1	Abbreviations	24
2.2	Analog inputs and binary inputs	12	4.2	Conversion tables	28
	2.2.1 Speed demand	12	4.3	MTU contact persons/service partners	32
	2.2.2 Engagement control	14			
	2.2.3 Trolling	16	5	Appendix B	
			5.1	Index	33

1 Safety

1.1 Important provisions for all products

Nameplate

The product is identified by nameplate, model designation or serial number and must match with the information on the title page of this manual.

Nameplate, model designation or serial number can be found on the product.

General information

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Modifications or conversions
- Noncompliance with the safety instructions and warning notices

Correct use

The product is intended exclusively for the application specified in the contract or defined at the time of delivery.

This means that the equipment must be operated:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/MTU contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (including engine control/parameters)
- In compliance with all safety regulations and in adherence with all warning notices in this manual
- In accordance with the maintenance requirements over the entire service life of the product (→ Maintenance Schedule)
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair
- By contracting only workshops authorized by the manufacturer to carry out repair and overhaul

Any other use is considered improper use and increases the risk of personnel injury or material damage in product operation. The manufacturer will accept no liability for such damage.

Modifications or conversions

Unauthorized modifications to the product compromise safety.

The manufacturer will accept no liability or warranty claims for any damage caused by unauthorized modifications or conversions.

Spare parts

Only genuine spare parts must be used to replace components or assemblies.

The manufacturer will accept no liability or warranty claims for any damage caused by the use of other spare parts.

1.2 Personnel and organizational requirements

Organizational measures of the operator

This manual must be issued to all personnel involved in operation, maintenance, repair or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair and transport personnel at all times.

Use this manual as a basis for instructing personnel on product operation and repair, whereby the safety-relevant instructions, in particular, must be read and understood.

This is particularly important in the case of personnel who only occasionally perform work on or around the product. This personnel must be instructed repeatedly.

Personnel requirements

All work on the product shall be carried out by trained and qualified personnel only.

- Training at the Training Center of the manufacturer
- Qualified personnel specialized in mechanical and plant engineering

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair and transport.

Working clothes and personal protective equipment

Wear proper protective clothing for all work.

When working, always wear the necessary personal protective equipment (e.g. ear protectors, protective gloves, goggles, breathing protection). Observe the information on personal protective equipment in the respective activity description.

1.3 Safety regulations for startup and operation

Safety regulations for startup

Install the product correctly and carry out acceptance in accordance the manufacturer's specifications before putting the product into service.

Before the product is put into operation for the first time, all official authorizations must be available and commissioning preconditions met.

When putting the product into operation, always ensure that

- All maintenance and repair work has been completed.
- All loose parts have been removed from rotating machine components.
- All personnel is clear of the danger zone surrounding moving parts of the machine.
- The service room is adequately ventilated.
- The exhaust system is leak-tight and that the gases are vented to atmosphere.
- Protect battery terminals, generator terminals or cables against accidental contact.

Immediately after putting the product into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems work properly.

Safety regulations for equipment operation

The operator must be familiar with the controls and displays.

The operator must be familiar with the consequences of any operations performed.

During operation, the display instruments and monitoring units must be permanently observed with regard to present operating status, violation of limit values and warning or alarm messages.

Malfunctions and emergency stop

The procedures for cases of emergency, in particular, emergency stop, must be practiced regularly.

The following steps must be taken if a malfunction of the system is recognized or reported by the system:

- Inform supervisor(s) in charge.
- Analyze the message.
- If required, carry out emergency operations e.g. emergency stop.

Operation

Do not remain in the operating room when the product is running for any longer than absolutely necessary.

Keep a safe distance away from the product if at all possible. Do not touch the product unless expressly instructed to do so following a written procedure.

Do not inhale the exhaust gases of the product.

The following requirements must be fulfilled before the product is started:

- Wear ear protectors.
- Mop up any leaked or spilled fluids and lubricants immediately or soak up with a suitable binder agent.

Operation of electrical equipment

When electrical equipment is in operation, certain components of these appliances are electrically live.

Observe the safety instructions for these devices.

1.4 Safety regulations for maintenance and repair work

Safety regulations prior to maintenance and repair work

Have maintenance or repair work carried out by qualified and authorized personnel only.

Allow the product to cool down to less than 50°C before starting maintenance work (risk of explosion of oil vapors, fluids and lubricants, risk of burning).

Before starting work, relieve pressure in systems and compressed-air lines which are to be opened. Use suitable containers of adequate capacity to catch fluids and lubricants.

When changing the oil or working on the fuel system, ensure that the service room is adequately ventilated.

Never carry out maintenance and repair work with the product in operation, unless:

- Expressly permitted to do so following a written procedure.
- The product is running in the low load range and only for as long as absolutely necessary.

Secure the product against unintentional starting, e.g. with start interlock.

Attach "Do not operate" sign in the operating area or to control equipment.

Disconnect the battery. Lock contactors.

Close the main valve on the compressed-air system and vent the compressed-air line when pneumatic starters are fitted.

Disconnect the control equipment from the product.

The following applies to starters with copper-beryllium alloy pinions:

- Wear breathing protection of filter class P3 during maintenance work. Do not blow out the interior of the flywheel housing or the starter with compressed air. Clean the flywheel housing inside with a class H dust extraction device.
- Observe the safety data sheet.

Safety regulations during maintenance and repair work

Take special care when removing ventilation or plug screws from the product. Cover the screw or plug with a rag to prevent fluids escaping under pressure.

Take care when draining hot fluids and lubricants (risk of burning).

Use only proper and calibrated tools. Observe the specified tightening torques during assembly or disassembly.

Carry out work only on assemblies or plants which are properly secured.

Never use lines for climbing.

Keep fuel injection lines and connections clean.

Always seal connections with caps or covers if a line is removed or opened.

Take care not to damage lines, in particular fuel lines, during maintenance and repair work.

Ensure that all retainers and dampers are installed correctly.

Ensure that all fuel injection and pressurized oil lines are installed with enough clearance to prevent contact with other components. Do not place fuel or oil lines near hot components.

Do not touch elastomeric seals if they have carbonized or resinous appearance unless hands are properly protected.

Note cooling time for components which are heated for installation or removal (risk of burning).

When working high on the equipment, always use suitable ladders and work platforms. Make sure components or assemblies are placed on stable surfaces.

Ensure particular cleanness during maintenance and repair work on the product. After completion of maintenance and repair work, make sure that no unattached parts are in/on the product (e.g. cloths and cable ties).

Safety regulations after completion of maintenance and repair work

Before barring, make sure that nobody is standing in the danger zone of the product.

Check that all guards have been reinstalled and that all tools and loose parts have been removed after working on the product (in particular, the barring tool).

Welding work

Welding operations on the product or mounted units are not permitted. Cover the product when welding in its vicinity.

Before starting welding work:

- Switch off the power supply master switch.
- Disconnect the battery.
- Separate the electrical ground of electronic equipment from the ground of the unit.

No other maintenance or repair work must be carried out in the vicinity of the product while welding is going on. Risk of explosion or fire due to oil vapors and highly flammable fluids and lubricants.

Do not use product as ground terminal.

Never position the welding power supply cable adjacent to, or crossing wiring harnesses of the product. The welding current may otherwise induce an interference voltage in the wiring harnesses which could conceivably damage the electrical system.

Remove parts (e.g. exhaust pipes) which are to be welded from the product beforehand.

Hydraulic installation and removal

Check satisfactory function and safe operating condition of tools, jigs and fixtures to be used. Use only the specified jigs and fixtures for hydraulic removal/installation procedures.

Observe the max. permissible force-on pressure specified for the jig/fixture.

Do not attempt to bend or apply force to lines.

Before starting work, pay attention to the following:

- Vent the hydraulic installation/removal jig, the pumps and the lines at the relevant points for the equipment to be used (e.g. open vent plugs, pump until bubble-free air emerges, close vent plugs).
- For hydraulic installation, screw on the jig with the piston retracted.
- For hydraulic removal, screw on the jig with the piston extended.

For a hydraulic installation/removal jig with central expansion pressure supply, screw spindle into shaft end until correct sealing is established.

During hydraulic installation and removal, ensure that nobody is standing in the immediate vicinity of the component to be installed/removed.

Working with batteries

Observe the safety instructions of the battery manufacturer when working with batteries.

Gases emanating from the battery are explosive. Avoid sparks and naked flames.

Do not allow electrolyte to come in contact with skin or clothing.

Wear goggles and protective gloves.

Never place tools on the battery.

Before connecting the cable to the battery, check the battery polarity. Battery pole reversal may lead to injury through the sudden discharge of acid or bursting of the battery body.

Working on electrical and electronic assemblies

Always obtain the permission of the person in charge before commencing maintenance and repair work or switching off any part of the electronic system required to do so.

De-energize the appropriate areas prior to working on assemblies.

Do not damage cabling during removal work. When reinstalling ensure that wiring is not damaged during operation by contact with sharp objects, by rubbing against other components or by a hot surface.

Do not secure cables on lines carrying fluids.

Do not use cable straps to secure cables.

Always use connector pliers to tighten union nuts on connectors.

Subject the device as well as the product to a function check on completion of all repair work. In particular, check the function of the engine emergency stop feature.

Store spare parts properly prior to replacement, i.e. protect them against moisture in particular. Pack defective electronic components and assemblies in a suitable manner when dispatched for repair, i.e. protected, in particular, against moisture and impact and wrapped in antistatic foil if necessary.

Working with laser equipment

When working with laser equipment, always wear special laser-protection goggles (hazard due to heavily focused radiation).

Laser equipment must be equipped with the protective devices necessary for safe operation according to type and application.

For conducting light-beam procedures and measurement work, only the following laser devices must be used:

- Laser devices of classes 1, 2 or 3A.
- Laser devices of class 3B, which have maximum output in the visible wavelength range (400 to 700 nm), a maximum output of 5 mW, and in which the beam axis and surface are designed to prevent any risk to the eyes.

1.5 Standards for safety notices in the text

DANGER 	In the event of immediate danger. Consequences: Death, serious or permanent injury! • Remedial action.
WARNING 	In the event of a situation involving potential danger. Consequences: Death, serious or permanent injury! • Remedial action.
CAUTION 	In the event of a situation involving potential danger. Consequences: Minor or moderate injuries! • Remedial action.
NOTICE 	In the event of a situation involving potentially adverse effects on the product. Consequences: Material damage! • Remedial action. • Additional product information.

Safety notices

- This manual with all safety instructions and safety notices must be issued to all personnel involved in operation, maintenance, repair or transportation.

2 Functional Description

2.1 Interfaces on LOP to non-MTU remote control systems

MTU provides an interface on the LOP for connection of a non-MTU remote control system (FRCS). Local Operating Panel LOP is the local control unit in the MTU automation system.

The following functions are available for the non-MTU remote control system:

Inputs on LOP

- Engine speed demand via analog signal (4 to 20 mA analog input)
- Engine speed demand via binary signals (Backup mode):
 - Engine speed increase
 - Engine speed decrease
- Engine speed demand switchover
- Engagement setting
- “Trolling ON” feedback (for alarm interlock)

Outputs on LOP

- Actual engine speed output
- Engine load signal output
- Command for emergency disengagement and speed reduction to idling
- Local mode status
- Engagement window status
- External engagement interlock status
- Engagement feedback

Power supply

The +24 VDC FRCS (external) and GND FRCS (external) signals must be provided by the non-MTU remote control system!

This allows the non-MTU remote control system to maintain control should the LOP fail when the engine is running, provided that power is still supplied to the EIM and Engine Control Unit.

Gear monitoring and gear control

The gear monitoring and control functions are available options for customer-supplied gearboxes.

2.2 Analog inputs and binary inputs

2.2.1 Speed demand

Two operating modes are possible – Automatic and Backup.

An 4 to 20 mA analog engine speed demand signal is expected in Automatic mode. Binary signals are expected in Backup mode by way of contrast. A binary input signal controls switching between these operating modes.

Analog speed demand (Automatic)

An analog signal (current) applied at this input is evaluated as a speed demand.

Input specification:

- Current:
 - 0 to 20 mA nom.
 - 0 to 25 mA max.
- Compliance voltage: 4.5 V at 20 mA
- Electrical isolation: 50 VDC
- Fault detection current input:
 - $I < 4$ mA
 - $I > 20$ mA

Connector	Pin	Signal	Description
X32	18	4 to 20 mA	4 mA corresponds to idling speed
X32	20	Return, GND isolated	20 mA corresponds to nominal engine speed

Table 1: Analog speed demand (Automatic)

Binary speed demand (Backup)

The “Engine speed increase” and “Engine speed decrease” inputs are only evaluated when the “Engine speed demand switchover” signal is active.

A signal briefly applied to these inputs increases or decreases the present engine speed by a certain value. The speed increases or decreases along a defined ramp if the signal is applied for a longer period.

Input specification:

- Switches: External voltage
- Input voltage 0 to 32 VDC
- LOW detection: $U_{in} < 4$ V
- HIGH detection: $U_{in} > 8$ V
- Input impedance: 12.1 k Ω
- Input filter: $f < 19$ Hz
- Electrical isolation: Up to 50 VDC

Connector	Pin	Signal	Description
X32	12	+24 VDC FRCS (external), max. 8 mA	Speed demand increase
X32	14	+24 VDC FRCS (external), max. 8 mA	Speed demand decrease
X32	9, 13, 17	GND FRCS (external)	Fuse 5 A

Table 2: Binary speed demand (Backup)

Engine speed demand switchover

This input switches over the speed demand modes (Automatic or Backup).

- Input permanently active → BACKUP setpoint signal (binary)
- Input inactive → AUTOMATIC (4 to 20 mA) active

Note:

- The previously set speed demand is maintained on switching over from AUTOMATIC -> BACKUP. The engine speed may be increased or decreased starting from this value by pressing the UP/DOWN keys respectively.
- The previously set speed demand is not maintained on switching over from BACKUP -> AUTOMATIC. The latest 4 to 20 mA speed demand takes immediate effect and is approached accordingly.

Input specification:

- Switches: External voltage
- Input voltage: 0 to 32 VDC
- LOW detection: $U_{in} < 4 \text{ V}$
- HIGH detection: $U_{in} > 8 \text{ V}$
- Input impedance: 12.1 k Ω
- Input filter: $f < 19 \text{ Hz}$
- Electrical isolation: Up to 50 VDC

The switching function must be checked manually at regular intervals.

Connector	Pin	Signal	Description
X32	10	+24 VDC FRCS (external), max. 8 mA	Speed demand switchover
X32	9, 13, 17	GND FRCS (external)	Fuse 5 A

Table 3: Engine speed demand switchover

2.2.2 Engagement control

In non-MTU remote control mode the engagement settings from the non-MTU remote control system are switched directly to the solenoid valves on the gearbox via relay contacts.

The MTU system has no influence on the engagement signals from the non-MTU remote control system in non-MTU remote control mode!

All engagement command signals shall be static. The maximum valve current is limited and protected by 3 A fuses. A valve current of 2 A should therefore not be exceeded.

Engagement control connection depends on the type of gearbox.

Reversing gearbox

The ahead valve of the gearbox must be wired up to the ahead output of the LOP. The astern valve of the gearbox must be wired up to the astern output of the LOP. All shafts are wired up in the same way in the case of standard cabling.

The valve which is activated for engagement in Local mode is set by software configuration (Configurator plug-in of the BlueVision New Generation Automation Suite commissioning tool). This is only relevant in the case of local gearbox control with LOP14.

This software configuration has no effect on valve control by the non-MTU remote control system! The engagement settings of the non-MTU remote control system must be assigned to the corresponding valve inputs by appropriately wiring up Yard slot connector X32 of the LOP (not at the gearbox connector!).

Example:

- The gearbox is activated correctly if the engagement input signal is connected to the ahead valve and the desired thrust is exerted at the propeller/impeller. The ahead/engagement input signal must be connected to astern if the propeller/impeller fails to exert the desired thrust.

Reduction gearbox

Reduction gearboxes only feature one engagement valve. This may be the ahead or astern valve depending on the gearbox concerned (manufacturer, model). The engagement valve must be wired up accordingly. The valve which is activated for engagement in Local mode is set by software configuration (Configurator plug-in of the BlueVision New Generation Automation Suite commissioning tool). This is only relevant in the case of local gearbox control with LOP14.

This software configuration has no effect on valve control by the non-MTU remote control system! The engagement settings of the non-MTU remote control system must be assigned to the corresponding valve inputs by appropriately wiring up Yard slot connector X32 of the LOP (not at the gearbox connector!).

Example:

- The gearbox is activated correctly if the engagement input signal is connected to the ahead valve and the gearbox engages. The engagement input signal must be connected to astern if gearbox fails to engage.

ATTENTION:

- The maximum admissible duration of the backflushing process for WaterJet gearboxes (WJ) is defined by the gearbox manufacturer. Operators shall observe this specification!

The "AL FRCS Ctrl. Fail" alarm is signaled and logged should the MTU system detect a difference between the engagement command from the non-MTU remote control system and activation of the valve at the gearbox connector.

Engagement setting ahead

Binary input for engagement setting ahead.

Connector	Pin	Signal	Description
X32	11	+24 VDC FRCS (external)	Engagement setting ahead, fuse 3 A
X32	9, 13, 17	GND FRCS (external)	Fuse 5 A

Table 4: Engagement setting ahead

Engagement setting neutral

Binary input for engagement setting neutral.

Connector	Pin	Signal	Description
X32	15	+24 VDC FRCS (external)	Engagement setting neutral, fuse 3 A
X32	9, 13, 17	GND FRCS (external)	Fuse 5 A

Table 5: Engagement setting neutral

Engagement setting astern

Binary input for engagement setting astern.

Connector	Pin	Signal	Description
X32	19	+24 VDC FRCS (external)	Engagement setting astern, fuse 3 A
X32	9, 13, 17	GND FRCS (external)	Fuse 5 A

Table 6: Engagement setting astern

2.2.3 Trolling

Trolling ON feedback

Binary input for Trolling ON feedback.

Gear control oil pressure monitoring is suppressed when the input is active.

Connector	Pin	Signal	Description
X32	16	+24 VDC (external), Low < 4 V; High > 8 V	Trolling ON feedback
X32	9, 13, 17	GND FRCS (external)	Fuse 5 A

Table 7: Trolling ON feedback

2.3 Analog outputs and binary outputs

2.3.1 Analog outputs

Engine speed

A DC voltage signal the value of which is proportional to the engine speed is issued at this output.

Connector	Pin	Signal	Description
X32	2	0 to 10 VDC, max. 8 mA, min. 1.25 kΩ, 50 V _{Iso}	0 rpm to 3000 rpm (programmable)
X32	6	GND	-

Table 8: Analog engine speed

Engine load

The engine load signal is issued at this output. The current torque is normalized to rated torque (rated torque = torque at rated speed and rated power).

The engine load signal is greater than 100% if the current torque at rated power is less than rated torque.

Connector	Pin	Signal	Description
X32	4	0 to 10 VDC, max. 8 mA, min. 1.25 kΩ, 50 V _{Iso}	0 to 120%
X32	8	GND	-

Table 9: Analog engine speed

2.3.2 Binary outputs

Note: The binary outputs do not feature open-line monitoring (wire break).

Engagement state feedback signals

The SPU provides binary outputs for engagement state feedback via floating relay contacts at connector X36.

Note:

- The signals must be assigned to the corresponding outputs by software configuration (Configurator plug-in of the BlueVision New Generation Automation Suite commissioning tool).

Engagement state ahead feedback

Engagement ahead feedback is issued at this output.

Connector	Pin	Signal	Description
X36	11	NO	Engagement ahead feedback
X36	7	NC	Engagement ahead feedback
X36	9	COM	Fuse 1 A

Table 10: Engagement state ahead feedback

Engagement state neutral feedback

Engagement neutral feedback is issued at this output.

Connector	Pin	Signal	Description
X36	5	NO	Engagement neutral feedback
X36	1	NC	Engagement neutral feedback
X36	3	COM	Fuse 1 A

Table 11: Engagement state neutral feedback

Engagement state astern feedback

Engagement astern feedback is issued at this output.

Connector	Pin	Signal	Description
X36	6	NO	Engagement astern feedback
X36	2	NC	Engagement astern feedback
X36	4	COM	Fuse 1 A

Table 12: Engagement state astern feedback

Forced/emergency disengagement and idling speed

The output is active when the forced disengagement command is applied (e.g. during engine start, stop and emergency stop).

This signal must reset the engagement control of the non-MTU remote control system to its initial state (i.e. gearbox in NEUTRAL or DISENGAGED).

This contact may also be used to set speed demand automatically to idling for increased operational safety. This ensures that the engine speed does not increase uncontrollably when started in non-MTU remote mode.

Connector	Pin	Signal	Description
X32	7	GND switching, max. 100 mA	Switching when forced/emergency disengagement command is active.
X32	9, 13, 17	GND FRCS (external)	-

Table 13: Forced/emergency disengagement and idling speed

Local mode

The output is active when Local mode is activated at the LOP (MTU-LOP control). When the plant is in Local mode it is controlled exclusively by the MTU-LOP. In this case the non-MTU remote control system is deactivated.

ATTENTION:

- Always check the engine speed and engagement state settings before switching between Local and Remote modes to avoid any undesired movement of the vessel.
For maximum operational safety: Set IDLING SPEED and ENGAGEMENT NEUTRAL!
- The engine speed and engagement state setpoint signals are always transferred and executed immediately on switching over to another operating mode.

Connector	Pin	Signal	Description
X32	3	GND switching, max. 100 mA	Switching when Local mode is active.
X32	9, 13, 17	GND FRCS (external)	-

Table 14: Local mode

Engagement window

The output is active when the engine is running outside the speed range in which the clutch can be actuated.

Connector	Pin	Signal	Description
X32	5	GND switching, max. 100 mA	Switching when engine speed is not in engagement window.
X32	9, 13, 17	GND FRCS (external)	-

Table 15: Engagement window

Engagement interlock

The output is active when the “External engagement interlock” input is active.

Connector	Pin	Signal	Description
X32	1	GND switching, max. 100 mA	Switching when the external “Engagement interlock” input is active.
X32	9, 13, 17	GND FRCS (external)	-

Table 16: Engagement interlock

3 Installation and Commissioning Instructions

3.1 Connecting a non-MTU remote control system (shipyard supply) to LOP

Preconditions

- ☑ Engine shut down and starting disabled.
- ☑ Operating voltage switched off at master switch (fuse).
- ☑ Cable conductor cross-sections for connection to Yard slot connectors have been checked (solid core and fine strand): 0.08 to 1.5 mm²; fine strand with wire end ferrule: 0.25 to 1.0 mm²).

Special tools, Material, Spare parts

Designation / Use	Part No.	Qty.
Torque wrench, 0.5-5 Nm	(→ Tools Catalog)	1

NOTICE



Short circuit.

Risk of malfunction or system failure!

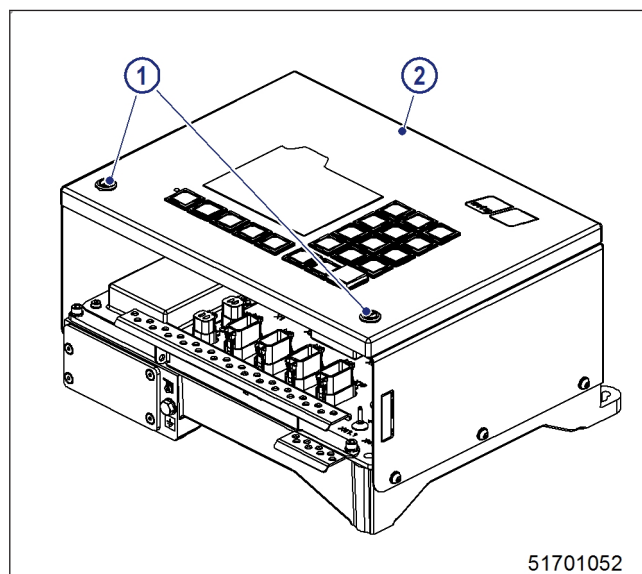
- Make sure that no parts fall into the casing.

Opening Local Operating Panel LOP

1. Release the catch (1) of the front panel (2) using a screwdriver.

Note: When open, the cover is higher than the top edge of the LOP.

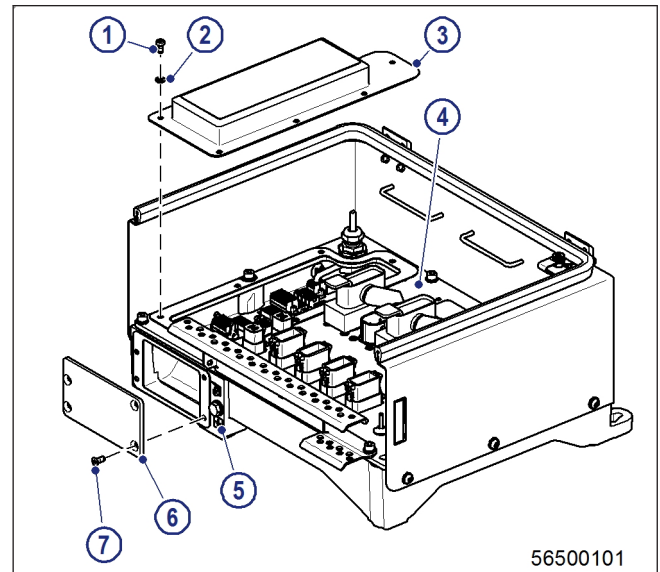
2. Slowly open up the front panel (2) as far as it will go.



51701052

Opening Yard slot / cable entry plate

1. Unscrew screws (1) and remove Yard slot cover (3) with washers (2) from SPU front panel (4).
2. Unscrew screws (7) and remove cable entry plate (6) together with routed cables from SPU (5).



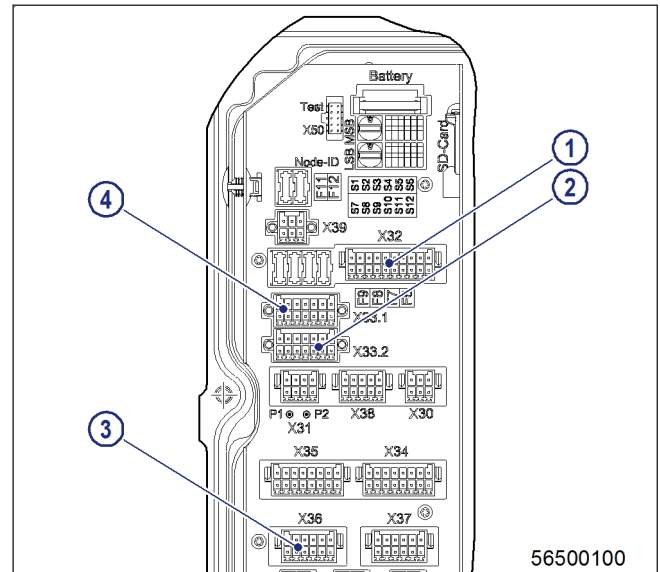
56500101

Installing cable at cable entry plate

- Note: The shipyard is responsible for routing cables in and applying the conductors to the connectors.
1. Install a number of cable bushings on the cable entry plate in accordance with the cabling scope.
 2. Install unions together with routed cables on the cable entry plate.

Cabling in Yard slot of LOP

- Note:
- The shipyard is responsible for routing cables in and applying the conductors to the connectors.
 - The connectors for the Yard slot are included in the scope of delivery.
 - The maximum cross-section for conductors at the connector contacts in the Yard slot is 1.5 mm².
1. Relocate jumper plug X33 from header X33.1 (4) to header X33.2 (2).



56500100

2. Connect non-MTU remote control system cable conductors to connector X32 (1).

Connector	Pin	Channel designation	Signal designation
X32	1	RO6M6	External engagement interlock (binary)
X32	2	AOU1M	Engine speed (0 to 10 V)
X32	3	RO7M7	Local mode (binary)
X32	4	AOU2M	Engine load (0 to 10 V)
X32	5	RO8M8	Engagement window (binary)
X32	6	AOU_GND	GND engine speed
X32	7	RO9M9	Forced disengagement command (binary)

Connector	Pin	Channel designation	Signal designation
X32	8	AOU_GND	GND engine load
X32	9	GND_FRCS	GND for binary outputs
X32	10	ERCS_CMD	Speed demand switchover (binary)
X32	11	ERCS_AH_M	Gearbox control ahead (binary)
X32	12	ERCS_RPM_UP	Speed increase (binary)
X32	13	GND_FRCS	GND for binary outputs
X32	14	ERCS_RPM_DOWN	Speed decrease (binary)
X32	15	ERCS_N_M	Gearbox control neutral (binary)
X32	16	SW1MR	Trolling ON feedback (binary)
X32	17	GND_FRCS	GND for binary outputs
X32	18	RPM_I_IN	Speed demand (4 to 20 mA)
X32	19	ERCS_AS_M	Gearbox control astern (binary)
X32	20	RPM_I_RETURN	RET/GND isolated speed demand (4 to 20 mA)

Table 17: Pin assignment, connector X32

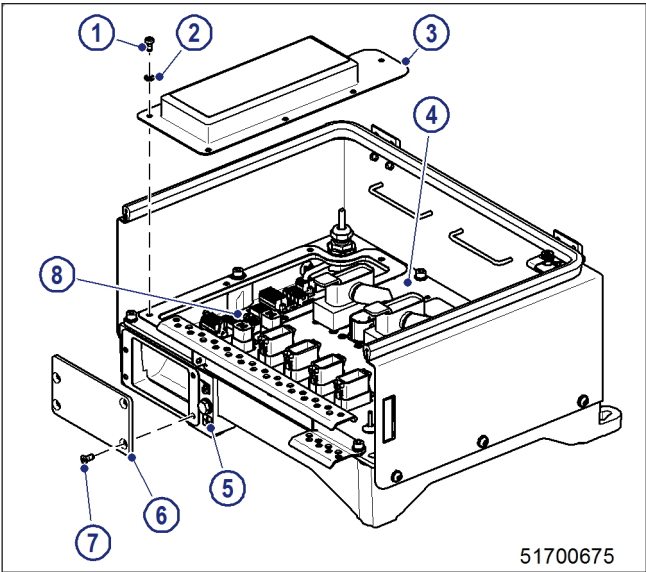
3. Connect engagement state feedback cable conductors to connector X36 (3).

Connector	Pin	Channel designation	Signal designation
X36	1	GRO2M2R2_NC	Gearbox feedback neutral / summary alarm (output)
X36	2	GRO3M3R3_NC	Gearbox feedback astern / speed window 2 (output)
X36	3	GRO2M2R2_COM	Gearbox feedback neutral / summary alarm (output)
X36	4	GRO3M3R3_COM	Gearbox feedback astern / speed window 2 (output)
X36	5	GRO2M2R2_NO	Gearbox feedback neutral / summary alarm (output)
X36	6	GRO3M3R3_NO	Gearbox feedback astern / speed window 2 (output)
X36	7	GRO1M1R1_NC	Gearbox feedback ahead / speed window 1 (output)
X36	8	Not used	
X36	9	GRO1M1R1_COM	Gearbox feedback ahead / speed window 1 (output)
X36	10	TOL3M3_GND	GND power supply
X36	11	GRO1M1R1_NO	Gearbox feedback ahead / speed window 1 (output)
X36	12	TOL3M3_OUT	24 V - power supply (max. 1 A)

Table 18: Pin assignment, connector X36

Closing Yard slot / cable entry plate

- 1. Mount cable entry plate (6) together with routed cables on SPU (5) using screws (7).



- 2. Tighten screws (7) to specified torque using a torque wrench.

Name	Size	Type	Lubricant	Value/Standard
Screw		Tightening torque		3 Nm

- 3. Check that all connectors in the Yard slot (8) are plugged in.
- 4. Install Yard slot cover (3) on SPU front panel (4) using screws (1) and washers (2).
- 5. Tighten screws (1) to specified torque using a torque wrench.

Name	Size	Type	Lubricant	Value/Standard
Screw		Tightening torque		3 Nm

Closing Local Operating Panel LOP 14

- For installation, follow reverse sequence of working steps.

4 Appendix A

4.1 Abbreviations

Abbrevia- tion	Meaning	Explanation
A/D	Analog/Digital converter	Converts sensor voltages into numeric values
ADEC	Advanced Diesel Engine Control	Engine governor
AGR	AbgasrückführungExhaust gas recirculation (EGR)	Exhaust gas recirculation (EGR)
AL	Alarm	General alarm
ANSI	American National Standards Institute	Association of American standardization organizations
ATL	Abgasturbolader	Exhaust turbocharger
BDM	Backup Data Module	Data storage module
BR	Baureihe	Series
BS	Base Station	
BV	Betriebsstoffvorschrift	Fluids and Lubricants Specifications, MTU Publication No. A01061/..
CAN	Controller Area Network	Data bus system, bus standard
CCP	Central Connecting Plate	Mounting plate
CCS	Compatibility Check System	
CCU	Central Control Unit	Central control unit of the SmartLine system
CDC	Calibration Drift Compensation	Drift compensation setting in the Engine Control Unit using DiaSys
CMPS	Compass	
CPP	Controllable Pitch Propeller	
CU	Control Unit	Central connection unit for the joystick system
°C	Degrees Celsius	
DBGI	DeBuG Interface	
DDV	Diagnose Daten Verbindung	Diagnostic data connection
DILA	Diagnostic lamp	
DIN	Deutsches Institut für Normung e. V.	German Standardization Organization, at the same time identifier of German standards ("Deutsche Industrie-Norm")
DIS	Display Unit	Display panel
DL	Default Lost	Alarm: CAN bus failure
DMC	Data Matrix Code	
DMC dec.	Data Matrix Code decoded	
DOC	Diesel Oxidation Catalyst	Oxidation catalyst upstream of diesel particulate filter
DPF	Diesel Particulate Filter	
DTS	Depot Test System	
ECS	Engine Control System	Engine management system
ECU	Engine Control Unit	Engine governor
EDM	Engine Data Module	Memory module for engine data

Abbrevia- tion	Meaning	Explanation
EEPROM	Electrically Erasable and Programmable Read Only Memory	Memory module
EIL	Engine Identification Label	Electronic nameplate which identifies the engine
EIM	Engine Interface Module	
EMU	Engine Monitoring Unit	
EPC	Electronic Pump Control	Pump control unit
EPROM	Electrically Programmable Read Only Memory	Memory module
ESD	Electrostatic Device	Component susceptible to electrostatic discharge
ESI	Emergency Stop Input	Emergency stop input at the engine governor
ETC	Electronic Transmission Controller	J1939 message
ETK	Ersatzteilkatalog	Spare parts catalog
FCU	Flap Control Unit	Exhaust flap control unit
FDH	Field Data Handling	
FPP	Fixed Pitch Propeller	
FW	Firmware	
FSW	Firm software	
GCU	Gear Control Unit	
GMU	Gear Monitoring Unit	
h	Hour	
HE	Haupteinspritzung	Main injection
HI	High	Alarm: Measured value exceeds 1st maximum limit value
HIHI	High High	Alarm: Measured value exceeds 2nd maximum limit value
HRC	Human Readable Code	
HT	High Temperature	
Hz	Hertz	
ICFN	ISO – Continuous rating – Fuel stop power – Net	Engine power rating as per DIN-ISO 3046-7
IDM	Interface Data Module	Memory module for interface data
IMO	International Maritime Organisation	
IP	International Protection	Degree of protection against accidental contact, dust and moisture as per DIN 40050
ISO	International Organization for Standardization	International umbrella organization of all national institutes of standardization
JSK	Joystick	
KBP	Kraftstoff Booster Pump	Fuel booster pump
kg	Kilogram	
kΩ	Kilo ohm	
KGS	Kraftgegenseite	Engine free end as per DIN ISO 1204
KJ	Key switch	
KS	Kraftseite	Engine driving end as per DIN ISO 1204

Abbrevia- tion	Meaning	Explanation
LCD	Liquid Crystal Display, Liquid Crystal Device	
LCU	Local Control Unit	LOP subassembly
LED	Light Emitting Diode	
LL	Low Limit	Lower limit value
L1L	Low 1 Limit	First lower limit value
L2L	Low 2 Limit	Second lower limit value
LMU	Local Monitoring Unit	LOP subassembly
LO	Low	Alarm: Measured value lower than 1st minimum limit value
LOLO	Low Low	Alarm: Measured value lower than 2nd minimum limit value
LOP	Local Operating Panel	Local control panel
LOS	Local Operating Station	Local control unit
LPU	Lubricating oil Pump control Unit	
MCD	Marine Control Drive	Remotely-controllable and regulated fluid coupling
MCS	Monitoring and Control System	
MCU	Monitoring Computer Unit	Management Computer Unit
MD	Missing Data	No data available
MEU	Monitoring Extension Unit	Extended engine and gearbox monitoring
MFD	Multifunction Display	
MG	Message	
MPC	MTU Personal Computer	
MPU	Microprocessor Unit, Microprocessing Unit	
mm	Millimeter	
MTU	Motoren- und Turbinen-Union	
MΩ	Mega ohm	
ms	Millisecond	
ND	Niederdruck	Low pressure
NE	Nacheinspritzung	Post-injection
NOx	Collective term for gaseous oxides of nitrogen	
sNE	späte Nacheinspritzung	Late post-injection
NT	Niedrigtemperatur	Low temperature
OT	Oberer Totpunkt	Top dead center
PROFET	Protected FET	
P-xyz	Pressure-xyz	Pressure measuring point, xyz specifies the measuring point designation
PAN	Panel	Control panel
PCS	Process Control System	
PCU	Propeller Control Unit	
PGN	Parameter Group Number	J1939 message
PIM	Peripheral Interface Module	

Abbrevia- tion	Meaning	Explanation
PIU		
PPC	Priming Pump Control	Priming pump controller
PPS	Programmable Process Station	
PSU		
PVC	Polyvinyl chloride	
RAM	Random Access Memory	
RCS	Remote Control System	
RL	Redundancy Lost	Alarm: Redundant CAN bus failure
RMD	Remote Maintenance and Diagnosis System	
RP	Rudder Propeller	
SAE	Society of Automotive Engineers	U.S. standardization organization
SAM	Service and Automation Module	
SD	Sensor Defect	Alarm: Sensor failure
SDAF	Shut Down Air Flaps	Emergency air-shutoff flap(s)
SE	System Error	Operating system error
SILA	Safety lamp	
SLD	Second Level Domain	
SMU	Shaft Speed Monitoring Unit	Shaft speed acquisition
SPN	Suspect Parameter Number	J1939 message
SS	Safety System	Safety system alarm
SSK	Schnellschlussklappe(n)	Emergency air-shutoff flap(s)
T-xyz	Temperature-xyz	Temperature measuring point, xyz specifies the measuring point designation
TD	Transmitter Deviation	Alarm: Deviation in transmitter values
TFT	Thin Film Transistor	
TH	Thruster Box	Thruster control unit
UL	Upper Limit	Upper limit value
U1L	Upper 1 Limit	First upper limit value
U2L	Upper 2 Limit	Second upper limit value
UT	Unterer Totpunkt	Bottom Dead Center
VE	Voreinspritzung	Pilot injection
VS	Voith Schneider	Voith Schneider drive
VSP	Voith Schneider Propeller	
WJ	Waterjet	Water jet drive
WZK	Werkzeugkatalog	Tool Catalog
X	X	Marking in table to identify a signal as binary or analog
ZKP	Zugehörigkeit-Kategorie-Parameter	Assignment category parameter, numbering scheme for the signals of the ADEC engine governor

4.2 Conversion tables

Length

Unit A	multiplied by factor	= Unit B
in	25.4	= mm
ft	0.3048	= m
yd	0.9144	= m
stat. mile	1.609	= km
Nm	1.852	= km
yd	3	= ft
yd	36	= in

Unit B	multiplied by factor	= Unit A
mm	0.03937	= in
m	3.281	= ft
km	0.6215	= stat. mile

Area

Unit A	multiplied by factor	= Unit B
in ²	645.16	= mm ²
ft ²	0.0929	= m ²
yd ²	0.8361	= m ²
stat. mile ²	2.59	= km ²

Unit B	multiplied by factor	Unit A
mm ²	0.00155	= in ²
m ²	10.764	= ft ²
m ²	1.1960	= yd ²
km ²	0.3861	= stat. mile ²

Volume

Unit A	multiplied by factor	= Unit B
in ³	16387	= mm ³
ft ³	0.02832	= m ³
yd ³	0.7646	= m ³
gallon (U.S.)	3.785	= dm ³
gallon (UK)	4.546	= dm ³

Unit B	multiplied by factor	= Unit A
cm ³	0.06102	= in ³
m ³	35.31	= ft ³
dm ³	0.2642	= gallon (U.S.)
dm ³	0.22	= gallon (UK)

Volumetric flow

Unit A	multiplied by factor	= Unit B
gal/min (GPM, U.S.)	3.79	= l/min
gal/min (GPM, U.S.)	0.134	= ft ³ /min (cfm)
ft ³ /min (cfm)	1.70	= m ³ /h
Unit B	multiplied by factor	= Unit A
l/min	0.264	= gal/min (U.S.)
ft ³ /min (cfm)	7.48	= gal/min (U.S.)
m ³ /h	0.589	= ft ³ /min (cfm)

Velocity

Unit A	multiplied by factor	= Unit B
ft/s	0.3048	= m/s
stat. mile/h (mph)	1.609	= km/h
knot (UK)	1.852	= km/h
Unit B	multiplied by factor	= Unit B
m/s	3.281	= ft/s
km/h	0.6214	= stat. mile/h (mph)
km/h	0.54	= knot (UK)

Mass

Unit A	multiplied by factor	= Unit B
lb	0.4536	= kg
oz	28.35	= g
ton (imp.)	1.016	= t
Unit B	multiplied by factor	= Unit A
g	0.03527	= oz
kg	2.205	= lb
t	0.9842	= ton (imp.)

Force

Unit A	multiplied by factor	= Unit B
lbf	0.4536	= kp
lbf	4.4482	= N
kp	9.80665	= N
Unit B	multiplied by factor	= Unit A
kp	2.205	= lbf
N	0.10197	= kp
N	0.2248	= lbf

Density

Unit A	multiplied by factor	= Unit B
slug/ft ³	515.4	= kg/m ³
Unit B	multiplied by factor	= Unit A
kg/m ³	0.00194	= slug/ft ³

Torque

Unit A	multiplied by factor	= Unit B
lbf ft	1.3558	= Nm
Unit B	multiplied by factor	= Unit A
Nm	0.7376	= lbf ft

Pressure

Unit A	multiplied by factor	= Unit B
lbf/in ² (psi)	703.1	= kp/m ² (mmH ₂ O)
lbf/in ² (psi)	0.06895	= bar
lbf/ft ² (psf)	47.88	= Pa
inHg	0.03386	= bar
inHg	345.3	= kp/m ² (mmH ₂ O)
Unit B	multiplied by factor	= Unit A
atm	760	= mmHg
atm	1.0133	= bar
atm	10332	= kp/m ² (mmH ₂ O)
atm	1.0332	= kp/cm ² (at)
atm	14.696	= lbf/in ² (psi)
bar	14.504	= lbf/in ² (psi)
Pa	0.0209	= lbf/ft ² (psf)

2nd moment of mass

Unit A	multiplied by factor	= Unit B
slug ft ²	1.3558	= kg m ²
Unit B	multiplied by factor	= Unit A
kg m ²	0.7376	= slug ft ²

Energy

Unit A	multiplied by factor	= Unit B
lbf ft	1.356	= J
kcal	4186.8	= J
BTU	1055	= J
CHU	1899	= J

Unit B	multiplied by factor	= Unit A
J	0.7376	= lbf ft
J	0.0002388	= kcal
J	0.0009478	= BTU
J	0.00052656	= CHU

Power

Unit A	multiplied by factor	= Unit B
PS	0.7355	= kW
HP	0.7457	= kW
BTU/s	1.054	= kW
kcal/h	1.163	= W
HP	550	= lbf ft/s

Unit B	multiplied by factor	= Unit A
kW	1.36	= PS
kW	1.341	= HP
kW	0.9487	= BTU/s
W	0.8598	= kcal/h
lbf ft/s	0.0018	= HP

Temperature

	Celsius	Kelvin
x °C	-	= x + 273.15 K
x K	= x - 273.15 °C	-
x °F	= 5/9(x - 32) °C	= 5/9(x - 32) + 273.15 K
x °R	= 5/4x °C	= (5/4x) + 273.15 K

	Fahrenheit	Réaumur
x °C	= 9/5x + 32 °F	= (4/5x) °R
x K	= 9/5(x - 273.15) + 32 °F	= 4/5(x - 273.15) °R
x °F	-	= 4/9(x - 32) °R
x °R	= (9/4x) + 32 °F	-

Fuel consumption

Unit A	multiplied by factor	= Unit B
mile/gal (US)	0.4251	= km/l
gal/mile (US)	2.3527	= l/km

Unit B	multiplied by factor	= Unit A
km/l	2.3527	= mile/gal (US)
l/km	0.4251	= gal/mile (US)

4.3 MTU contact persons/service partners

Our worldwide sales network with its subsidiaries, sales offices, representatives and customer service centers ensures fast and direct support on site and the high availability of our products.

Local support

Experienced and qualified specialists place their knowledge and expertise at your disposal.

For locally available support, go to the MTU Internet site: <http://www.mtu-online.com>

24h hotline

With our 24h hotline and the outstanding flexibility of our service staff, we are always ready to assist you – either during operation, for preventive maintenance, corrective work in case of malfunction or changed operating conditions, or for spare parts supply.

Your contact person in our Customer Assistance Center:

E-mail: info@mtu-online.com

Tel.: +49 7541 9077777

Fax: +49 7541 9077778

Asia/Pacific: +65 6100 2688

North and Latin America: +1 248 560 8000

Spare parts service

Fast, simple and correct identification of spare parts for your drive system or vehicle fleet. The right spare part at the right time at the right place.

With this aim in mind, we can call on a globally networked spares logistics system, a central warehouse at headquarters and on-site stores at our subsidiary companies, agencies and service workshops.

Your contact at Headquarters:

E-mail: spare.parts@mtu-online.com

Tel.: +49 7541 908555

Fax: +49 7541 908121

5 Appendix B

5.1 Index

A

Abbreviations 24
Analog outputs 17

B

Binary outputs
- Engagement interlock 18
- Engagement state feedback 18
- Engagement window 18
- Forced/emergency disengagement and idling speed 18
- Local mode 18

C

Connecting a non-MTU remote control system (shipyard supply) to LOP 20
Contact persons 32
Conversion tables 28

E

Engagement control
- Engagement setting ahead 14
- Engagement setting astern 14
- Engagement setting neutral 14
- Reduction gearbox 14
- Reversing gearbox 14
Engine load 17
Engine speed 17

H

Hotline 32

I

Interface on LOP to non-MTU remote control systems
- Inputs on LOP 11
- Outputs on LOP 11
- Power supply 11

M

MTU contact persons 32

S

Safety notices, standards 10
Safety regulations
- Important provisions 4
- Maintenance work 7
- Operation 6
- Organizational requirements 5
- Personnel requirements 5
- Repair work 7
- Safety notices, standards 10
- Startup 6

Service partners 32

Spare parts service 32

Speed demand

- Analog speed demand (Automatic) 12
- Binary speed demand (Backup) 12
- Engine speed demand switchover 12

T

Trolling

- Trolling ON feedback 16